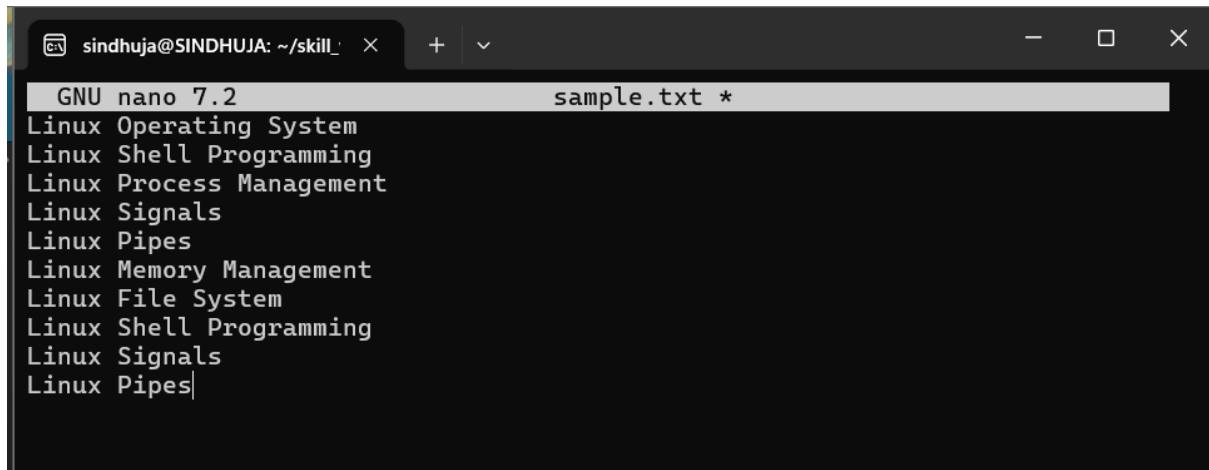


SKILL WEEK 12

S23: Question1

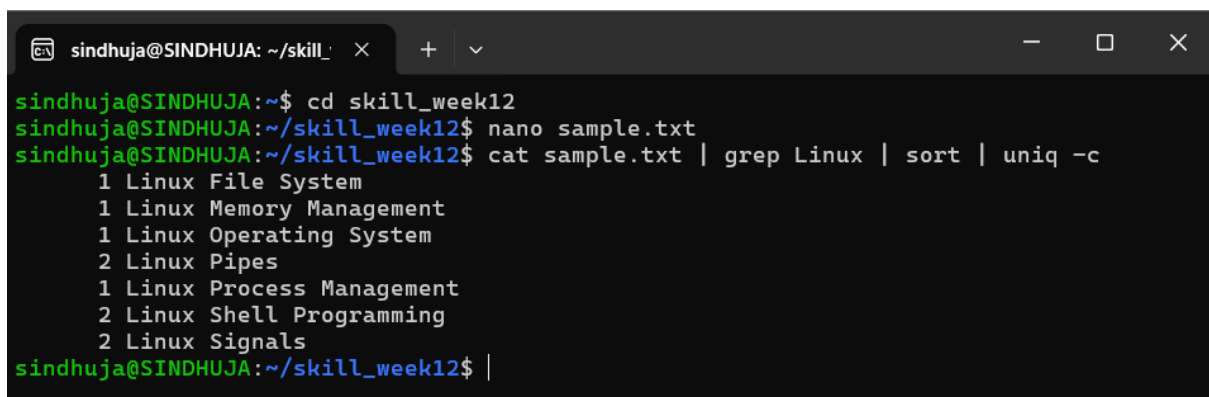
Aim:

To execute Linux pipelines, launch multiple background jobs, monitor running processes and system resources, detect job failures, and analyze performance using Linux commands.



```
GNU nano 7.2 sample.txt *
Linux Operating System
Linux Shell Programming
Linux Process Management
Linux Signals
Linux Pipes
Linux Memory Management
Linux File System
Linux Shell Programming
Linux Signals
Linux Pipes
```

Output:



```
sindhuja@SINDHUJA:~$ cd skill_week12
sindhuja@SINDHUJA:~/skill_week12$ nano sample.txt
sindhuja@SINDHUJA:~/skill_week12$ cat sample.txt | grep Linux | sort | uniq -c
      1 Linux File System
      1 Linux Memory Management
      1 Linux Operating System
      2 Linux Pipes
      1 Linux Process Management
      2 Linux Shell Programming
      2 Linux Signals
sindhuja@SINDHUJA:~/skill_week12$ |
```

```
sindhuja@SINDHUJA: ~/skill_ x + v
2 Linux Signals
sindhuja@SINDHUJA:~/skill_week12$ sleep 200 &
[1] 864
sindhuja@SINDHUJA:~/skill_week12$ sleep 250 &
[2] 899
sindhuja@SINDHUJA:~/skill_week12$ sleep 300 &
[3] 925
sindhuja@SINDHUJA:~/skill_week12$ jobs -l
[1] 864 Running sleep 200 &
[2]- 899 Running sleep 250 &
[3]+ 925 Running sleep 300 &
sindhuja@SINDHUJA:~/skill_week12$ ps -f
UID PID PPID C STIME TTY TIME CMD
sindhuja 293 292 0 05:52 pts/0 00:00:00 -bash
sindhuja 864 293 0 05:53 pts/0 00:00:00 sleep 200
sindhuja 899 293 0 05:53 pts/0 00:00:00 sleep 250
sindhuja 925 293 0 05:53 pts/0 00:00:00 sleep 300
sindhuja 1107 293 0 05:54 pts/0 00:00:00 ps -f
sindhuja@SINDHUJA:~/skill_week12$ top
top - 05:54:55 up 26 min, 1 user, load average: 0.01, 0.01, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.2 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 7782.6 total, 7227.0 free, 515.6 used, 190.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7267.0 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   21532  12960  9632  S   0.7   0.2   0:01.06 systemd
   177 message+  20   0   9644   5372  4656  S   0.3   0.1   0:00.15 dbus-daemon
   184 root        20   0   17968  8776   7732  S   0.3   0.1   0:00.07 systemd-lo+
  1154 sindhuja    20   0   9308   5688   3492  R   0.3   0.1   0:00.02 top
     2 root        20   0   3180   2212  2072  S   0.0   0.0   0:00.00 init-syste+
     6 root        20   0   3196   2132  2008  S   0.0   0.0   0:00.00 init
    52 root        19  -1  50324 15888 14724  S   0.0   0.2   0:00.37 systemd-jo+
   101 root        20   0  25404  6820  5156  S   0.0   0.1   0:00.11 systemd-ud+
   111 systemd+   20   0  21460 13264 11004  S   0.0   0.2   0:00.07 systemd-re+
   130 systemd+  20   0  91028  7960  6996  S   0.0   0.1   0:00.05 systemd-ti+
   176 root        20   0   4244   2676  2424  S   0.0   0.0   0:00.00 cron
   204 syslog     20   0  222516  5620  4552  S   0.0   0.1   0:00.09 rsyslogd
   207 root        20   0   3124   1976  1836  S   0.0   0.0   0:00.00agetty
   208 root        20   0 107020 23192 13732  S   0.0   0.3   0:00.08 unattended+
   291 root        20   0   3184   1116   980  S   0.0   0.0   0:00.00 SessionLea+
   292 root        20   0   3200   1252  1108  S   0.0   0.0   0:00.01 Relay(293)
   293 sindhuja    20   0   6080   5444  3700  S   0.0   0.1   0:00.04 bash
   294 root        20   0   6700   4668  3884  S   0.0   0.1   0:00.00 login
   339 sindhuja    20   0  20312 11616  9496  S   0.0   0.1   0:00.07 systemd
   340 sindhuja    20   0  21160  3600  1872  S   0.0   0.0   0:00.00 (sd-pam)
   362 sindhuja    20   0   6080   5356  3664  S   0.0   0.1   0:00.01 bash
   864 sindhuja    20   0   3132   1908  1792  S   0.0   0.0   0:00.00 sleep
   899 sindhuja    20   0   3132   1908  1792  S   0.0   0.0   0:00.00 sleep
   925 sindhuja    20   0   3132   1896  1784  S   0.0   0.0   0:00.00 sleep

sindhuja@SINDHUJA:~/skill_week12$ kill %2
sindhuja@SINDHUJA:~/skill_week12$ jobs -l
[1] 864 Running sleep 200 &
[2]- 899 Terminated sleep 250
[3]+ 925 Running sleep 300 &
sindhuja@SINDHUJA:~/skill_week12$ time ls -R
.:
report.txt skill12_task1 skill12_task2 test_cases.sh
sample.txt skill12_task1.c skill12_task2.c

real    0m0.004s
user    0m0.003s
sys      0m0.000s
sindhuja@SINDHUJA:~/skill_week12$ |
```

Observation:

1. Linux pipelines processed data through multiple commands.
2. Multiple background jobs were executed simultaneously.
3. Running processes were monitored using ps -f.
4. System resource usage was monitored using top.
5. A background job was terminated using kill.

6. Command execution time was measured using time.

Result:

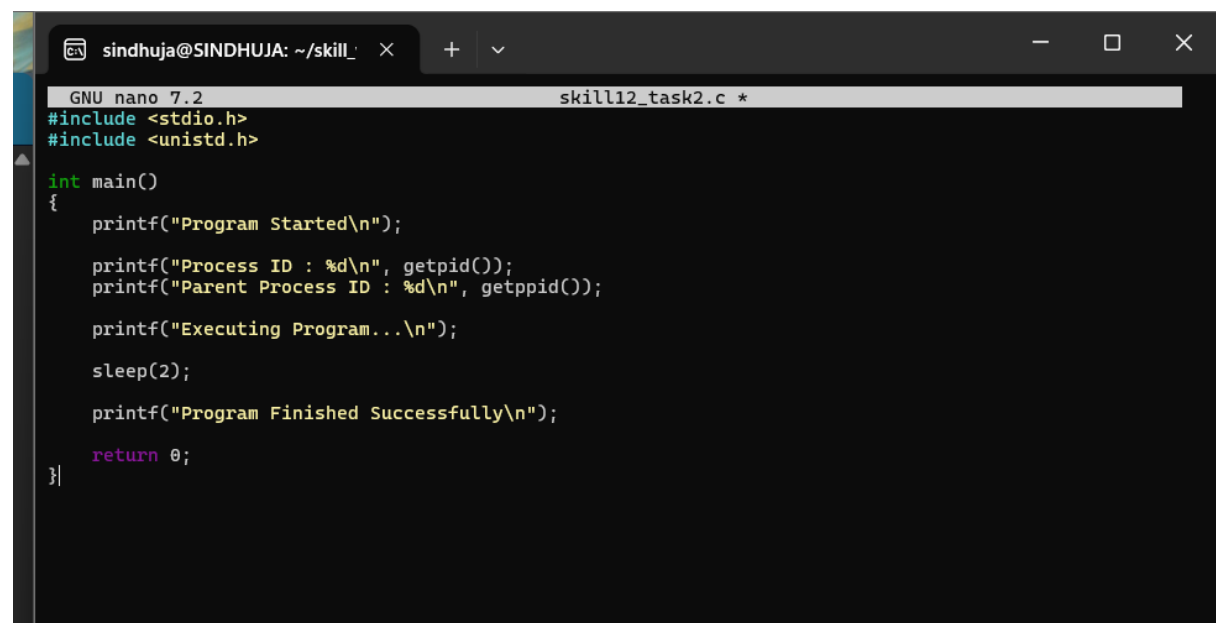
Successfully executed Linux pipelines, launched multiple background jobs, monitored running processes and system resources, detected job termination, and analyzed command performance using Linux commands.

S23: Question2

Aim:

To document the architecture of a Linux program, explain its execution path and system calls, create a flow diagram, record design decisions, and review the program documentation.

Program:

A screenshot of a terminal window with a nano editor. The window title is 'sindhuja@SINDHUJA: ~/skill_'. The editor shows the source code for 'skill12_task2.c'. The code includes <stdio.h> and <unistd.h>, and contains a main function that prints the program start, process ID, parent process ID, execution start, a 2-second sleep, and program finish.

```
GNU nano 7.2 skill12_task2.c *
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Program Started\n");

    printf("Process ID : %d\n", getpid());
    printf("Parent Process ID : %d\n", getppid());

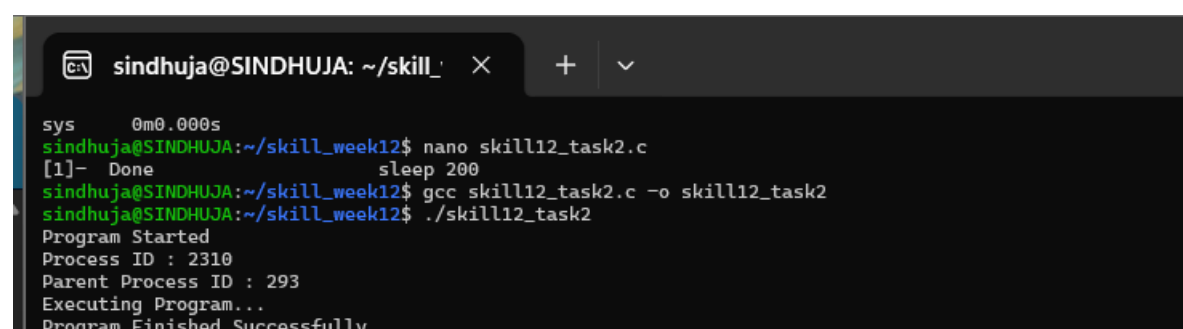
    printf("Executing Program...\n");

    sleep(2);

    printf("Program Finished Successfully\n");

    return 0;
}
```

Output:

A screenshot of a terminal window showing the execution of the program. The user runs 'nano skill12_task2.c', then 'gcc skill12_task2.c -o skill12_task2', and finally './skill12_task2'. The output matches the program's print statements.

```
sys      0m0.000s
sindhuja@SINDHUJA:~/skill_week12$ nano skill12_task2.c
[1]-  Done                  sleep 200
sindhuja@SINDHUJA:~/skill_week12$ gcc skill12_task2.c -o skill12_task2
sindhuja@SINDHUJA:~/skill_week12$ ./skill12_task2
Program Started
Process ID : 2310
Parent Process ID : 293
Executing Program...
Program Finished Successfully
```


1. getpid() returned the current process ID.
2. getppid() returned the parent process ID.
3. strace displayed the sequence of system calls made by the program.
4. The program executed successfully and terminated using exit_group().

Result:

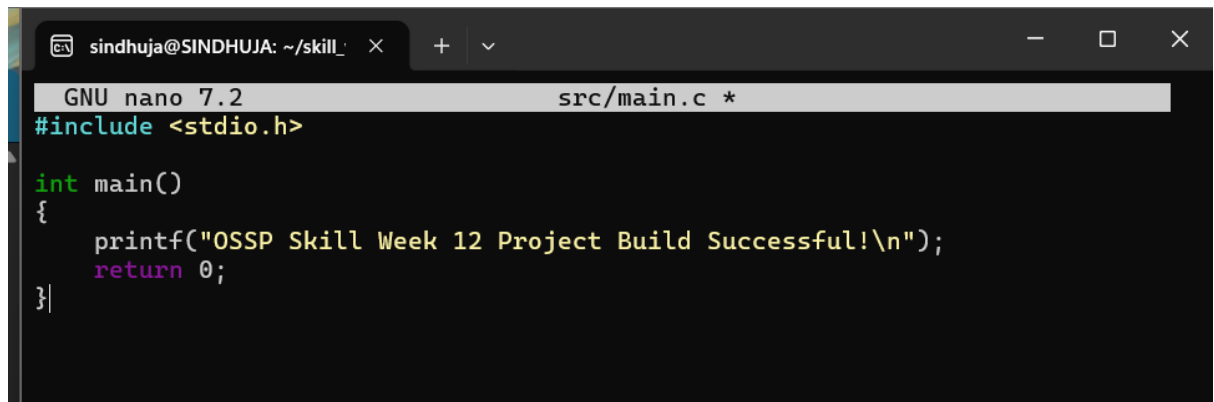
Successfully documented the program architecture, explained the execution path, identified system calls using strace, and verified the program execution flow in Linux.

S24: Question1

Aim:

To organize source code into separate folders, create a build script, add a README file, verify the repository structure, and validate the project deliverables in Linux.

Program:



```
GNU nano 7.2 src/main.c *
#include <stdio.h>

int main()
{
    printf("OSSP Skill Week 12 Project Build Successful!\n");
    return 0;
}
```

```
sindhuja@SINDHUJA: ~/skill_ x + v
GNU nano 7.2 include/calculator.h *
#ifndef CALCULATOR_H
#define CALCULATOR_H

void displayMessage();

#endif
```

```
sindhuja@SINDHUJA: ~/skill_ x + v
GNU nano 7.2 docs/README.md *
# OSSP Skill Week 12

## Project Structure
- src : Source code
- include : Header files
- docs : Project documentation

## Build Command
gcc src/main.c -o project_build

## Run Command
./project_build
```

```
sindhuja@SINDHUJA: ~/skill_ x + v
GNU nano 7.2 build.sh *
#!/bin/bash

echo "Building OSSP Project..."

gcc src/main.c -o project_build

echo "Build Completed Successfully."
```

Output:

```
sindhuja@SINDHUJA: ~/skill_ × + v
sindhuja@SINDHUJA:~$ cd skill_week12
sindhuja@SINDHUJA:~/skill_week12$ mkdir src include docs
sindhuja@SINDHUJA:~/skill_week12$ touch src/main.c
sindhuja@SINDHUJA:~/skill_week12$ touch include/calculator.h
sindhuja@SINDHUJA:~/skill_week12$ touch docs/README.md
sindhuja@SINDHUJA:~/skill_week12$ ls -R
.:
docs      report.txt  skill12_task1  skill12_task2  src
include   sample.txt  skill12_task1.c skill12_task2.c test_cases.sh

./docs:
README.md

./include:
calculator.h

./src:
main.c
sindhuja@SINDHUJA:~/skill_week12$ nano src/main.c
sindhuja@SINDHUJA:~/skill_week12$ nano include/calculator.h
sindhuja@SINDHUJA:~/skill_week12$ nano docs/README.md
sindhuja@SINDHUJA:~/skill_week12$ nano build.sh
sindhuja@SINDHUJA:~/skill_week12$ chmod +x build.sh
sindhuja@SINDHUJA:~/skill_week12$ ./build.sh
Building OSSP Project...
Build Completed Successfully.
sindhuja@SINDHUJA:~/skill_week12$ ./project_build
OSSP Skill Week 12 Project Build Successful!
sindhuja@SINDHUJA:~/skill_week12$ tree
.
├── build.sh
├── docs
│   └── README.md
├── include
│   └── calculator.h
├── project_build
├── report.txt
├── sample.txt
├── skill12_task1
├── skill12_task1.c
├── skill12_task2
├── skill12_task2.c
├── src
│   └── main.c
└── test_cases.sh

4 directories, 12 files
sindhuja@SINDHUJA:~/skill_week12$ |
```

Observation:

1. The project was organized into separate folders.
2. A README file documented the project.
3. A build script automated compilation.
4. The project compiled and executed successfully.
5. The repository structure was verified using the tree command.

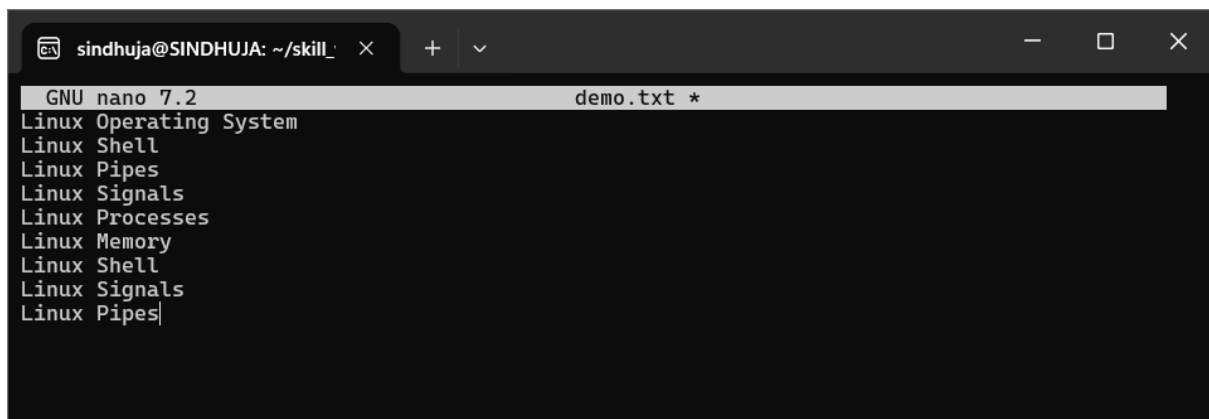
Result:

Successfully organized the source code, created a build script, added a README file, verified the repository structure, and validated the project deliverables in Linux.

S24: Question2

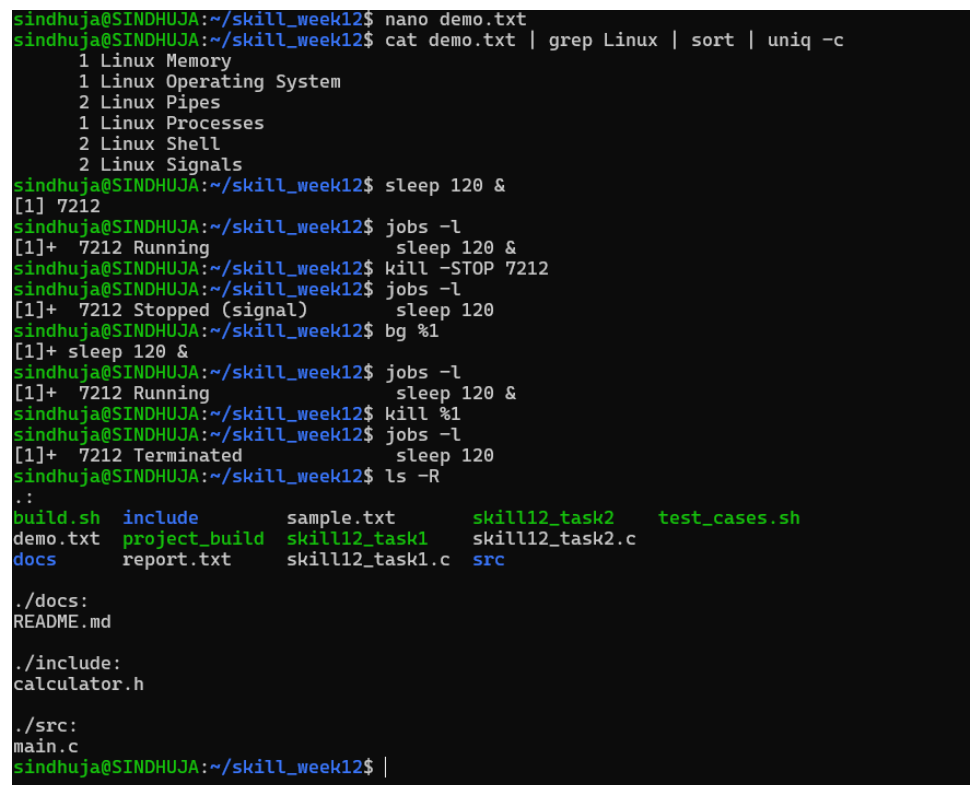
Aim:

To create demo scenarios, validate Linux features, demonstrate pipelines and signals, verify program execution, and perform a final project review in Linux.



```
sindhuja@SINDHUJA: ~/skill_ x + v - □ ×
GNU nano 7.2 demo.txt *
Linux Operating System
Linux Shell
Linux Pipes
Linux Signals
Linux Processes
Linux Memory
Linux Shell
Linux Signals
Linux Pipes|
```

Output:



```
sindhuja@SINDHUJA:~/skill_week12$ nano demo.txt
sindhuja@SINDHUJA:~/skill_week12$ cat demo.txt | grep Linux | sort | uniq -c
  1 Linux Memory
  1 Linux Operating System
  2 Linux Pipes
  1 Linux Processes
  2 Linux Shell
  2 Linux Signals
sindhuja@SINDHUJA:~/skill_week12$ sleep 120 &
[1] 7212
sindhuja@SINDHUJA:~/skill_week12$ jobs -l
[1]+  7212 Running                  sleep 120 &
sindhuja@SINDHUJA:~/skill_week12$ kill -STOP 7212
sindhuja@SINDHUJA:~/skill_week12$ jobs -l
[1]+  7212 Stopped (signal)         sleep 120
sindhuja@SINDHUJA:~/skill_week12$ bg %1
[1]+  sleep 120 &
sindhuja@SINDHUJA:~/skill_week12$ jobs -l
[1]+  7212 Running                  sleep 120 &
sindhuja@SINDHUJA:~/skill_week12$ kill %1
sindhuja@SINDHUJA:~/skill_week12$ jobs -l
[1]+  7212 Terminated             sleep 120
sindhuja@SINDHUJA:~/skill_week12$ ls -R
.:
build.sh  include  sample.txt  skill12_task2  test_cases.sh
demo.txt  project_build  skill12_task1  skill12_task2.c
docs      report.txt  skill12_task1.c  src
./docs:
README.md
./include:
calculator.h
./src:
main.c
sindhuja@SINDHUJA:~/skill_week12$ |
```

Observation:

1. A background process was created using the sleep command.
2. The process was suspended using the SIGSTOP signal.
3. The suspended process was resumed using the bg command.
4. The process was terminated successfully using the kill command.
5. The project directory structure was verified using ls -R.

Result:

Successfully created demo scenarios, validated Linux process management, demonstrated background job control using signals, and completed the final project review in Linux.